

Terminal Velocity of Single Drops in Stagnant Liquids*

Win MYINT**, Shigeo HOSOKAWA** and Akio TOMIYAMA***

** Kobe University, Graduate School of Science and Technology

1-1 Rokkodai, Nada, Kobe 657-8501, Japan

*** Kobe University, Faculty of Engineering

1-1 Rokkodai, Nada, Kobe 657-8501, Japan

E-mail: tomiyama@mech.kobe-u.ac.jp

Abstract

Terminal velocities of single drops rising through infinite stagnant liquids under wide ranges of fluid properties were measured to examine the effects of initial shape deformation, surfactants and the viscosity ratio. As a result, the following conclusions were obtained: (1) the terminal velocity V_T of a drop is not affected by initial shape deformation due to strong viscous damping of shape oscillation, (2) the drop drag coefficient C_D is a function of the drop Reynolds number Re , the viscosity ratio κ and surfactant concentration, (3) surfactants increase C_D and the influence of surfactants on C_D becomes weaker as κ increases, which is consistent with the Levich's drag model, (4) the effect of κ on C_D can be evaluated by the factor in the Levich's model, $[(2+3\kappa)/(1+\kappa)]$, even at intermediate drop Re numbers, and (5) the combination of the Levich's model and Schiller & Nauman's correlation gives good estimation of C_D of single drops in clean and fully contaminated systems under the conditions of $-11.6 < \log_{10} M < -0.9$, $1.7 \times 10^{-1} < Re < 2 \times 10^2$, $1.7 \times 10^{-2} < Eo < 12.1$ and $0.1 < \kappa < 100$.

Key words : Drop, Terminal Velocity, Viscosity Ratio, Drag Coefficient, Surfactant

1. Introduction

Sufficient knowledge on terminal velocities V_T of single fluid particles rising or falling through infinite stagnant liquids is required for modeling and simulation of dispersed two-phase flows in various industrial fields. Many researches⁽¹⁾⁻⁽¹¹⁾ on V_T of single fluid particles or solid particles in stagnant liquids have been conducted and reliable models and correlations for V_T or the drag coefficient C_D are available. Grace et al.⁽¹⁾ proposed a well-known graphical correlation among the Eötvös number Eo , Reynolds number Re and Morton number M for single fluid particles in infinite stagnant liquids. Here, Eo is the ratio of buoyant force F_b to interfacial tension force F_s , Re is the ratio of inertial force F_i to viscous force F_μ and M represents the fluid properties of two fluids. They are defined by

$$Eo = \frac{F_b}{F_s} = \frac{(\rho_c - \rho_d)gd^2}{\sigma} \quad (1)$$

$$Re = \frac{F_i}{F_\mu} = \frac{\rho_c V_T d}{\mu_c} \quad (2)$$

$$M = \frac{F_\mu^4 F_b}{F_s^3 F_i^2} = \frac{g\mu_c^4 (\rho_c - \rho_d)}{\rho_c^2 \sigma^3} \quad (3)$$

where d is the sphere-volume equivalent diameter of the fluid particle, g the acceleration due to gravity, ρ the density, μ the viscosity and σ the interfacial tension. The subscripts c and d denote the continuous and dispersed phases, respectively. Bond and Newton⁽²⁾ pointed out that small fluid particles with high viscosity behave as solid particles and the drag forces acting

on them follow the Stokes's law⁽³⁾. Hadamard⁽⁴⁾ and Rybczynski⁽⁵⁾ deduced a theoretical C_D model for single spherical fluid particles at very low Reynolds numbers as a function of the viscosity ratio $\kappa (= \mu_d/\mu_c)$:

$$C_D = \frac{8}{Re} \left(\frac{2 + 3\kappa}{1 + \kappa} \right) \quad (4)$$

For bubbles, the viscosity ratio is close to zero ($\kappa \cong 0$). For rigid particles, it becomes infinity ($\kappa \cong \infty$) and Eq. (4) reduces to the Stokes's drag model⁽³⁾. Levich⁽⁶⁾ proposed a C_D model which takes into account the effects of not only κ but also surfactants:

$$C_D = \frac{8}{Re} \left[\frac{2 + 3\kappa + \frac{3C}{\mu}}{1 + \kappa + \frac{C}{\mu}} \right] \quad (5)$$

where C is a coefficient expressing the retardation of interface motion by surfactants. This equation is applicable only to spherical fluid particles at extremely low Reynolds numbers ($Re < 1$).

As for bubbles at medium or high Reynolds numbers, Tomiyama et al.⁽¹⁰⁾ proposed C_D correlations for single bubbles rising through clean, slightly contaminated and fully contaminated liquids:

(a) For a clean system

$$C_D = \max \left\{ \min \left[\frac{16}{Re} \left(1 + 0.15Re^{0.687} \right), \frac{48}{Re} \right], \frac{8}{3} \frac{Eo}{Eo + 4} \right\} \quad (6)$$

(b) For a slightly contaminated system

$$C_D = \max \left\{ \min \left[\frac{24}{Re} \left(1 + 0.15Re^{0.687} \right), \frac{72}{Re} \right], \frac{8}{3} \frac{Eo}{Eo + 4} \right\} \quad (7)$$

(c) For a fully contaminated system

$$C_D = \max \left\{ \left[\frac{24}{Re} \left(1 + 0.15Re^{0.687} \right) \right], \frac{8}{3} \frac{Eo}{Eo + 4} \right\} \quad (8)$$

They also pointed out that V_T of a bubble depends on the bubble shape, which is strongly affected by initial shape deformation⁽¹¹⁾. However, there are few studies on V_T of a drop at intermediate and high Reynolds numbers.

In this study, terminal velocities of single drops rising through infinite stagnant liquids under a wide range of Re ($0.17 < Re < 200$) were measured to investigate the influences of surfactants and viscosity ratio κ on C_D and V_T of a drop. The effect of initial shape deformation on V_T was also discussed.

2. Experimental

Experiments were conducted at room temperature and atmospheric pressure. Figure 1 shows the schematic of experimental apparatus. The acrylic tank and the measurement system were the same as those used in our previous experiments on bubble terminal velocity⁽¹¹⁾. The width, depth and height of the tank were 0.20, 0.20 and 0.69 m, which were large enough to avoid the wall effects⁽⁹⁾. Silicon oils (Sin-etsu Silicone, KF96-30 and KF96-100) and glycerol-water solutions were used for the dispersed and continuous phases, respectively. Experimental conditions are summarized in Table 1. The concentration of glycerol-water solution was changed to examine the effects of the viscosity ratio ($\kappa = 0.1 - 100$) and the Morton number ($\log_{10} M = -0.9 - -11.6$) on V_T . Under a constant Morton number, the Eötvös number Eo was varied by changing the drop diameter. Various nozzles with different inner diameters ($D_N = 1.2 - 8.0$ mm) were used to alter the degree of initial shape deformation (large initial shape deformation occurs when d is larger than D_N). Triton X-100

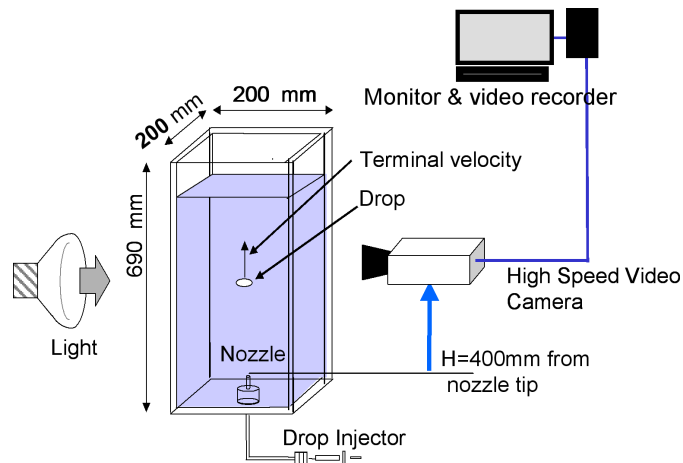


Fig. 1 Schematic of experimental apparatus

Table 1 Experimental conditions

Fluid (continuous phase)	Fluid (dispersed phase)	$\log_{10} M$	κ	ρ_c (kg/m ³)	ρ_d (kg/m ³)	μ_c (m.Pa.s)	μ_d (m.Pa.s)	σ (N/m)
Water (clean)	KF-30	-11.6	31.5	998	960	1	31.5	0.0546
Water (clean)	KF-100	-11.6	100	998	967	1	100	0.0508
Water (contaminated)	KF-30	-10.6	31.5	998	960	1	31.5	0.0246
Glycerol-water solution (clean)	KF-30	-6.0	1.9	1178	960	17	31.5	0.05
Glycerol-water solution (clean)	KF-30	-4.8	0.9	1198	960	34	31.5	0.0496
Glycerol-water solution (contaminated)	KF-30	-4.1	0.9	1198	960	34	31.5	0.0288
Glycerol-water solution (clean)	KF-30	-3.0	0.4	1221	960	87	31.5	0.0467
Glycerol-water solution (clean)	KF-100	-3.0	1.1	1223	969	89	100	0.0466
Glycerol-water solution (contaminated)	KF-30	-3.0	0.4	1221	960	87	31.5	0.0456
Glycerol-water solution (clean)	KF-30	-0.9	0.1	1246	960	280	31.5	0.0455

(C₈H₁₇C₆H₄(OCH₂CH₂)₁₀OH) was used for surfactants to examine its effects on V_T . The concentration of Triton X-100 was $2.68 \times 10^{-1} \text{ mol/m}^3$, which was four times higher than the maximum concentration in the research by Zhang and Finch⁽¹²⁾ ($6.7 \times 10^{-2} \text{ mol/m}^3$) and about a half of the critical micelle concentration ($4.2 \times 10^{-1} \text{ mol/m}^3$)⁽¹³⁾. Therefore, the contaminated systems in the present experiments could be regarded as fully contaminated systems and the influence of micelle was negligible.

Since all single drops in the present experiments rose rectilinearly and their shapes were axisymmetric, measurements of V_T , d and the aspect ratio E (the ratio of minor to major axes) were carried out using a single high-speed video camera (Kodak Motion Corder Analyzer SR-1000, shutter speed = 1/5000 s, frame rate = 250 frame/s, 512×480 pixels). Images for measuring V_T were taken at 400 mm above the nozzle tip, at which all drops reached terminal conditions. The spatial resolution of images ranged from 0.02 to 0.18 mm/pixel. The maximum error in the V_T measurement was 0.5% and the uncertainty in measured d estimated at 95 % confidence was 0.81 %. To examine the influence of initial shape deformation, rise velocity V_Z and aspect ratio E of a drop were measured near the nozzle tip using the high-speed video camera (Fig. 2). The liquid densities, viscosities and interfacial tension σ were

measured by a densimeter (JIS B7525, accuracy: $\pm 0.1\%$), a viscometer (Rion, Viscotester VT-03E, accuracy: $\pm 3\%$) and capillary tubes (glass, inner diameter: 0.63, 1.02 and 1.40 mm), respectively. The measurements of ρ_c , ρ_d , μ_c , μ_d and σ were carried out at least three times before and after each experiment. Uncertainties estimated at 95% confidence were 2.0, 3.0 and 2.0 % for ρ , μ and σ , respectively. The temperature of the continuous phase was monitored by a digital thermometer (Sato, SK-1250MC, accuracy: ± 0.53 K) and it was within the range of 293 – 295 K through all the experiments.

3. Results and discussions

The effect of initial shape deformation on V_T of single drops was examined and compared with that of single bubbles⁽¹⁴⁾. Figure 2 shows recorded images, measured aspect ratio E and rise velocities V_Z of single drops and bubbles of $Eu \cong 1.0$ released from nozzles with different diameters D_N . Bubbles or drops released from smaller diameter nozzles exhibit deformed initial shapes as shown in Fig. 2 (a) and (c), whereas the initial deformation of bubbles or drops released from larger diameter nozzles is small as shown in Fig. 2 (b) and (d). As for single bubbles, a slight difference in initial shape deformation causes a large difference in E and V_Z ⁽¹¹⁾. To the contrary, V_Z and E of single drops are not affected by initial shape deformation. This might be due to the higher viscous damping of shape oscillation of single drops than that of single bubbles. In fact, the strong fluctuation of E just after the release is observed only for bubbles as shown in Fig. 2 (c). Measured drop Reynolds numbers in clean and contaminated systems at $\log_{10} M = -3.0$ are plotted against Eu in Fig. 3 (a). For comparison, measured bubble Reynolds numbers at the same Morton number⁽¹⁴⁾ are shown in

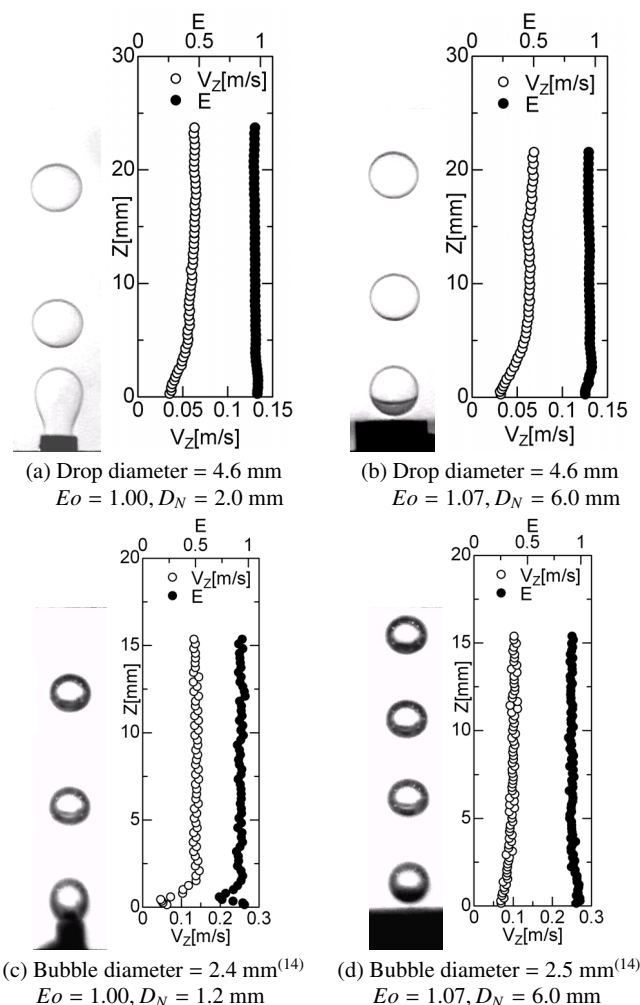


Fig. 2 Effects of initial shape deformation ($\log_{10} M \cong -4.8$ and $Eu \cong 1.0$)

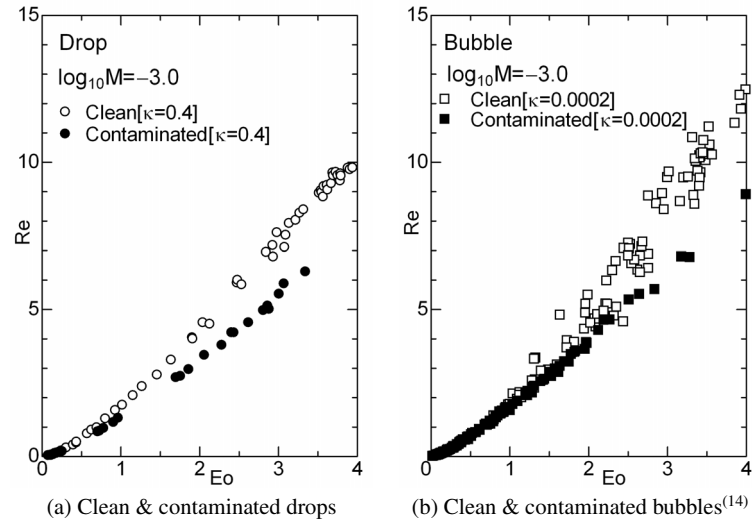


Fig. 3 Relation between measured Re and Eo for single bubbles and drops ($\log_{10} M = -3.0$)

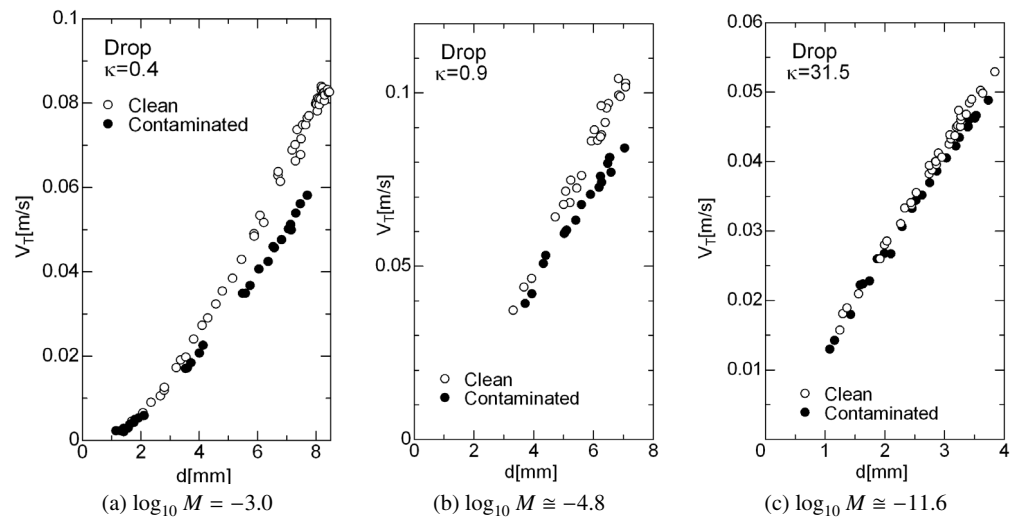


Fig. 4 Measured V_T in clean and contaminated systems

Fig. 3 (b). There is a large scatter of bubble Reynolds number in the clean system due to the difference in initial shape deformation, and the scatter in the contaminated system is very small because of the high damping of shape oscillation caused by surfactants⁽¹¹⁾. On the other hand, no scatter of drop Reynolds numbers is observed not only in the contaminated systems but also in the clean systems. These experimental data imply that there is little influence of initial shape deformation on V_T of single drops for clean and contaminated systems, and therefore, V_T and C_D are uniquely determined by fluid properties, d and surfactant concentration.

Comparisons of measured V_T of single drops between clean and contaminated systems are shown in Fig. 4. The V_T of single contaminated drops is lower than that of single clean drops. The difference in V_T between clean and contaminated systems becomes small in a high κ system as shown in Fig. 4 (c). Though the addition of surfactants changes σ , this change does not cause any change in V_T because the terminal condition in the present experimental conditions is determined by the viscous and inertial forces (viscous force dominating regime), in other words, by the drop Reynolds number. Measured C_D is plotted against Re in Fig. 5, which clearly indicates that C_D does not depend on M but on Re . As the viscosity ratio κ becomes smaller, C_D of contaminated drops becomes higher than that of clean drops. This increase of C_D must be due to the well-known Marangoni effect caused by the concentration gradient of adsorbed surfactants on the interface. To the contrary, the difference in C_D between

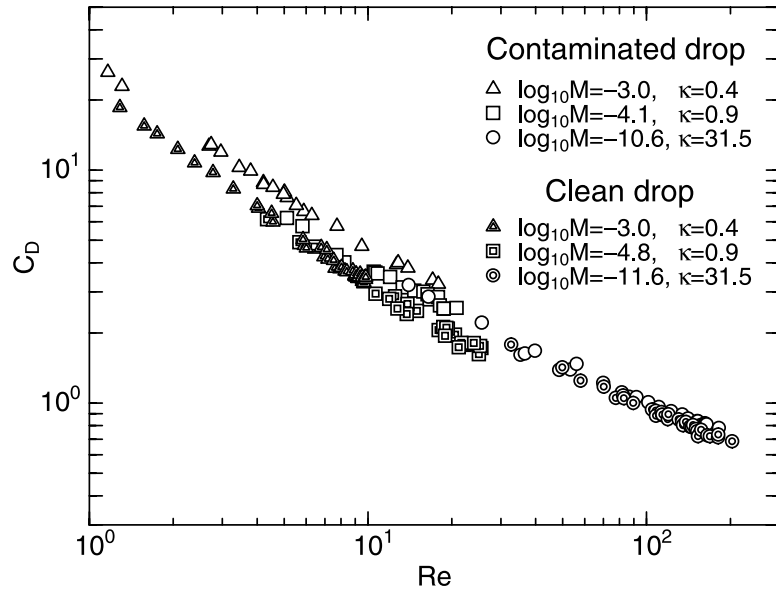


Fig. 5 Measured C_D and Re in clean and contaminated systems

the clean and contaminated systems becomes smaller as κ increases. According to Eq. (5), C_D of a spherical drop at a low Re number depends on $(\kappa + C/\mu)$, and therefore, the influence of surfactants on C_D and V_T becomes weaker as κ becomes larger and larger than C/μ . Hence, the experimental results indicate that the Levich's term $[(2 + 3\kappa + 3C/\mu)/(1 + \kappa + C/\mu)]$ in Eq. (5) might be applicable to C_D and V_T models not only for low Re number conditions but also for intermediate Re number conditions.

Figure 6 shows measured Re numbers of single clean drops with Eqs. (6) and (8) for bubbles in pure and fully contaminated systems under various conditions; $-11.6 < \log_{10} M < -0.9$, $0.1 < \kappa < 31.5$, $1.7 \times 10^{-1} < Re < 2 \times 10^2$ and $1.7 \times 10^{-2} < Eo < 12.1$. Drop shapes are also shown (the discussion on drop shapes will be reported in elsewhere). The measured Re under the largest κ condition ($\kappa = 31.5$) is close to Eq. (8) for contaminated bubbles. As κ decreases, the measured Re approaches to Eq. (6) for clean bubbles. This indicates that κ strongly affects the drop Re number, and that drop Re numbers would lie in the region between the curves of Eqs.(6) and (8). The first term in the right-hand side of Eq. (6) or Eq. (8) can be obtained by setting $(\kappa, C/\mu) = (0,0)$ or $(\kappa, C/\mu) = (0,\infty)$ in the following equation:

$$C_D = \frac{8(2 + 3\kappa + 3C/\mu)}{Re(1 + \kappa + C/\mu)}(1 + 0.15Re^{0.687}) \quad (9)$$

The above equation is the combination of Eq. (5) and the following equation proposed by Schiller and Nauman⁽¹⁵⁾ for single spherical solid particles.

$$C_D = \frac{24}{Re}(1 + 0.15Re^{0.687}) \quad (10)$$

For $(\kappa, C/\mu) = (\infty, 0)$, the right side of Eq. (9) becomes equal to the first term of Eq. (8), i.e., Eq. (10). Hence, the experimental results shown in Fig. 6 indicate that (i) the values of C_D for single drops are in the range between the values of C_D for clean bubbles and for solid particles, and (ii) Eq. (9) will be applicable to the evaluation of C_D for single drops under various κ conditions.

To examine the applicability of Eq. (9) to clean drops ($C = 0$) under various κ conditions, $C_D[(1 + \kappa)/8(2 + 3\kappa)]$ for clean drops for various systems is plotted against Re and compared with Eq. (9) in Fig. 7. There is a strong correlation between $C_D[(1 + \kappa)/8(2 + 3\kappa)]$ and Re .

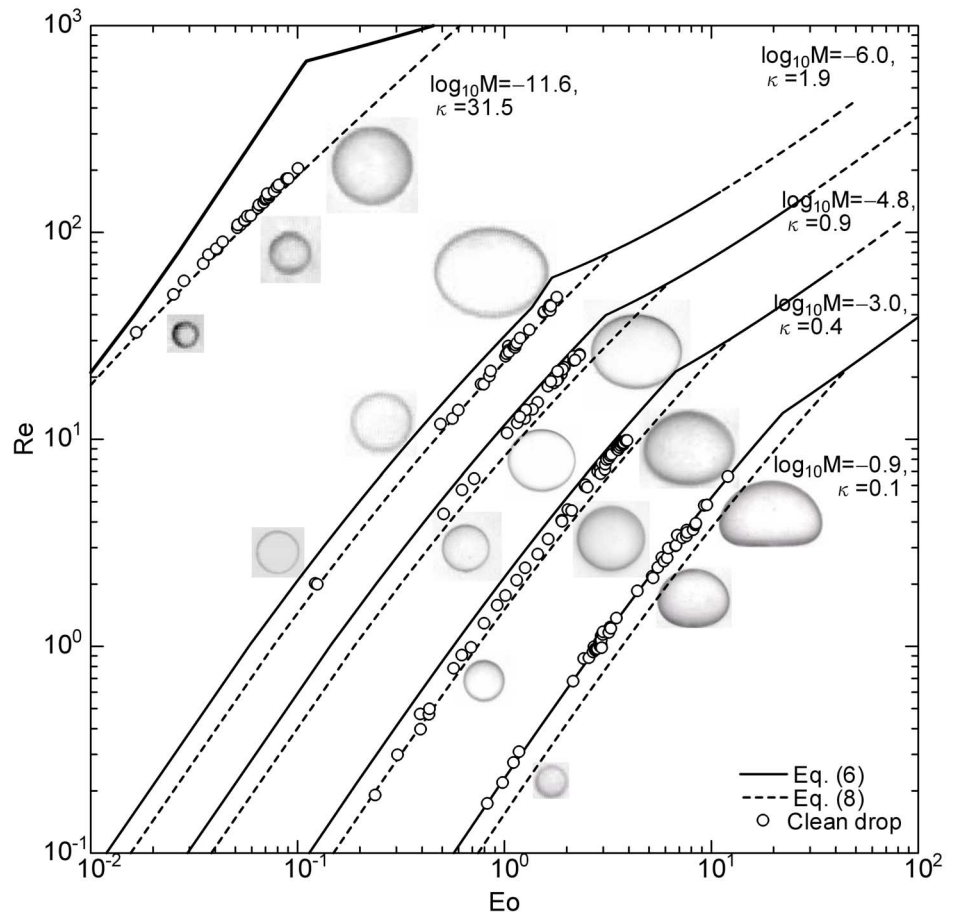


Fig. 6 Measured Re for single clean drops in various systems

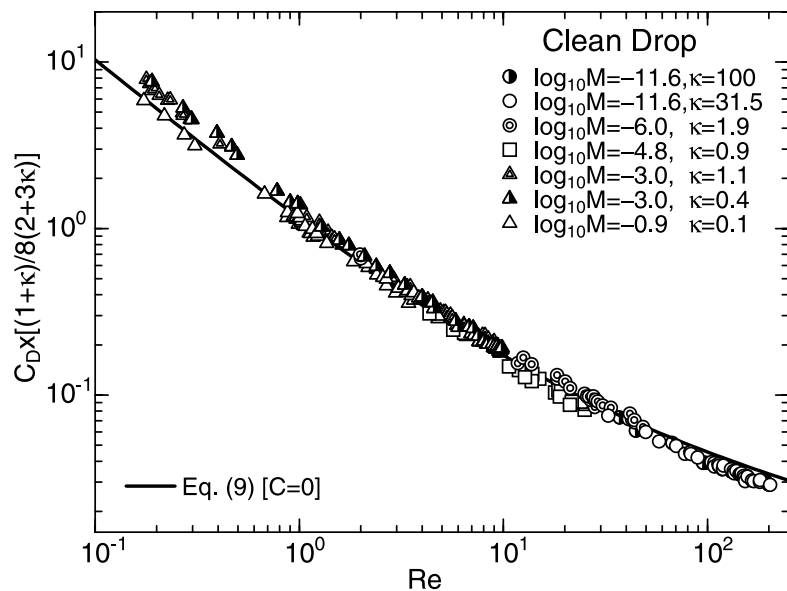


Fig. 7 Relation between measured C_D and Re of clean drops in various systems

The calculated C_D using Eq. (9) with $C = 0$ agrees well with the measured C_D of single clean drops. This result supports the speculation that the influence of κ on C_D for single drops can be evaluated by the term $[8(2 + 3\kappa)/(1 + \kappa)]$ not only in low Re but also in intermediate Re conditions ($Re < 200$).

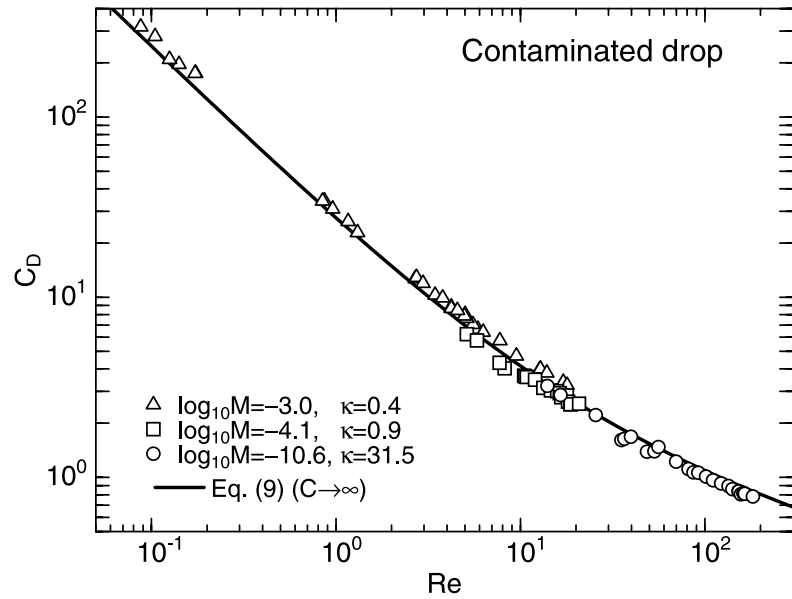


Fig. 8 Relation between measured C_D and Re of fully contaminated drops in various systems

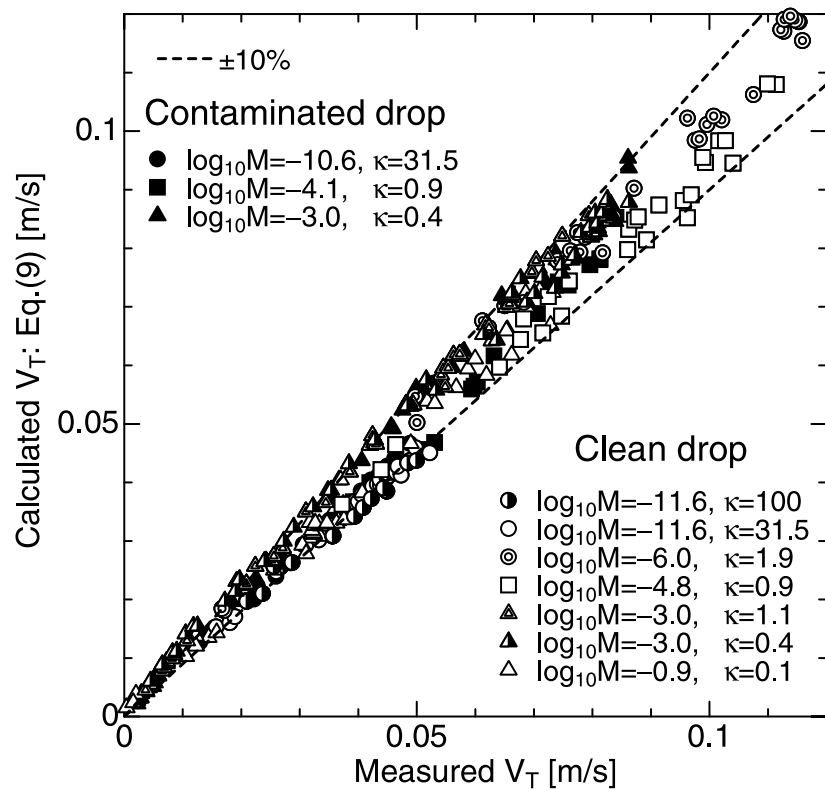


Fig. 9 Comparison between all the measured and calculated terminal velocities

Then, measured drag coefficients C_D of single drops in fully contaminated systems ($C \rightarrow \infty$) are plotted against Re and compared with Eq. (9) in Fig. 8. The measured C_D agrees well with C_D calculated by Eq. (9) with $C \rightarrow \infty$. This indicates that the influence of κ on V_T is negligible in fully contaminated systems as Eq. (9) implies. Comparison between all the measured and predicted terminal velocities are shown in Fig. 9. The calculated V_T agrees well with the measured V_T and the maximum error is about 10% under the conditions of $-11.6 < \log_{10} M < -0.9$, $1.7 \times 10^{-1} < Re < 2 \times 10^2$, $1.7 \times 10^{-2} < Eo < 12.1$ and $0.1 < \kappa < 100$.

4. Conclusion

Terminal velocities of single drops rising through infinite stagnant liquids under wide ranges of fluid properties were measured to examine the effects of initial shape deformation, surfactants and the viscosity ratio. As a result, the following conclusions were obtained.

- (1) The terminal velocity V_T of a drop is not affected by initial shape deformation due to strong viscous damping of shape oscillation.
- (2) The drop drag coefficient C_D is a function of the drop Reynolds number Re , the viscosity ratio κ and surfactant concentration.
- (3) Surfactants increase C_D and the influence of surfactants on C_D becomes weaker as κ increases. This trend is consistent with the Levich's drag model.
- (4) The effect of κ on C_D can be evaluated by the factor in the Levich's model, $[(2 + 3\kappa)/(1 + \kappa)]$, even at intermediate drop Re numbers.
- (5) The combination of the Levich's model and Schiller & Nauman's correlation gives good estimation of C_D of single drops in clean and fully contaminated systems under the conditions of $-11.6 < \log_{10} M < -0.9$, $1.7 \times 10^{-1} < Re < 2 \times 10^2$, $1.7 \times 10^{-2} < Eo < 12.1$ and $0.1 < \kappa < 100$.

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